

Chemistry · increasing difficulty · key terms, careful calculations and small details · Units 1, 2 and 3 (60/20/20)

Time: 65 minutes. **Total:** 80 points. Part A: 18 multiple-choice questions, 2 points each (36). Part B: 5 free-response questions (44). Weighting: Unit 1 60%, Unit 2 20%, Unit 3 20%. The questions get harder as you go, and this exam is marked strictly: a missing charge, unit, lone pair or key term loses the point.

Reference. Average atomic mass = sum of (isotope mass $\times$ fractional abundance). Electronegativity: H 2.1 $\cdot$ C 2.5 $\cdot$ N 3.0 $\cdot$ O 3.5 $\cdot$ F 4.0 $\cdot$ Na 0.9 $\cdot$ Mg 1.2 $\cdot$ Al 1.5 $\cdot$ Si 1.8 $\cdot$ P 2.1 $\cdot$ S 2.5 $\cdot$ Cl 3.0 $\cdot$ K 0.8 $\cdot$ Ca 1.0 $\cdot$ Br 2.8. Bond type from Δ EN: below 0.5 nonpolar covalent $\cdot$ 0.5 to 1.7 polar covalent $\cdot$ above 1.7 ionic. A periodic table and a calculator may be used.

LEVEL 1 · WARM-UP

- Chemistry Interim · Practice Exam 2 (Challenge)

7. How many electrons are in one ${}^{56}_{26}\text{Fe}^{3+}$ ion?

- 8.** Which pair are isotopes of the same element?

9. Bromine has two isotopes, Br-79 and Br-81. Its average atomic mass is 79.90 amu. Which statement is true?

- A** Br-79 is far more abundant.
B The two isotopes are present in nearly equal amounts.
C Br-81 is far more abundant.
D Nothing can be said without the exact masses.

10. Which configuration shows an atom in an excited state?

- A** $1s^2 2s^2 2p^6$ **B** $1s^2 2s^2 2p^6 3s^2$
C $1s^2 2s^2 2p^5 3s^1$ **D** $1s^2 2s^2 2p^6 3s^2 3p^1$

11. Which of these has the smallest radius?

- A** O^{2-} **B** F^-
C Na^+ **D** Mg^{2+}

12. Which formula is paired with its correct name?

- A** N_2O_3 , dinitrogen trioxide **B** CuCl_2 , copper chloride
C CaCl_2 , calcium dichloride **D** Na_2SO_4 , sodium sulfide

13. What is the noble gas configuration of selenium ($Z = 34$)?

- A** [Ar] 4s² 4p⁴ **B** [Kr] 4s² 3d¹⁰ 4p⁴
- C** [Ar] 4s² 3d¹⁰ 4p⁴ **D** [Ar] 3d¹⁰ 4p⁶

14. In a hydrogen atom, which transition emits the photon with the highest energy?

- A** $n = 4$ to $n = 3$ **B** $n = 3$ to $n = 2$
C $n = 4$ to $n = 2$ **D** $n = 2$ to $n = 1$

15. Which particle completes this fusion equation? ${}^2_1\text{H} + {}^2_1\text{H} \rightarrow {}^3_1\text{H} + ?$

- A** $\frac{1}{0}\text{n}$
- B** $\frac{1}{1}\text{H}$
- C** $\frac{4}{2}\text{He}$
- D** $\frac{0}{-1}\text{e}$

16. The first four ionization energies of an element are 578, 1817, 2745 and 11 577 kJ/mol. The element is most likely in group

- A** 1 **B** 2
C 13 **D** 14

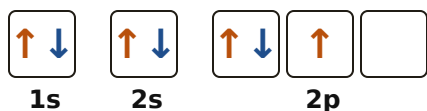
- 17.** Why is the first ionization energy of chlorine higher than that of sodium?
- A** Chlorine has a greater nuclear charge with about the same shielding, so its outer electrons are held more tightly.
 - B** Chlorine has more energy levels.
 - C** Chlorine's outer electrons are more shielded.
 - D** Chlorine is a larger atom.
- 18.** Which molecule has polar bonds but is nonpolar overall?
- A** H₂O
 - B** NH₃
 - C** Cl₂
 - D** CF₄

Part B · Free Response (show your work)

19. Find the mistakes (8 points) Unit 1

A student wrote the four answers below. Each one contains at least one mistake. For each, state every mistake and write the correct answer. (2 points each)

A. Orbital diagram of nitrogen (Z = 7)



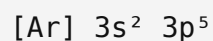
B. Electron configuration of iron (Z = 26)



C. Symbol for an ion with 12 protons, 13 neutrons and 10 electrons



D. Noble gas configuration of chlorine (Z = 17)



A. Mistake(s) and correct answer:

B. Mistake(s) and correct answer:

C. Mistake(s) and correct answer:

D. Mistake(s) and correct answer:

20. Careful calculations (10 points) Unit 1

(a) Complete the table. (3 points)

Species	Protons	Neutrons	Electrons
$^{31}_{15}\text{P}^{3-}$			
$^{65}_{29}\text{Cu}^{2+}$			
$^{137}_{56}\text{Ba}^{2+}$			

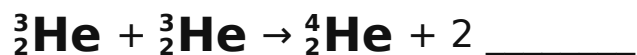
(b) Copper has two isotopes. Cu-63 has a mass of 62.930 amu and makes up 69.15% of copper. The average atomic mass of copper is 63.546 amu. Calculate the mass of the other isotope, Cu-65. (3 points)

Work:

(c) Lithium has two isotopes: Li-6 (6.015 amu) and Li-7 (7.016 amu). Its average atomic mass is 6.940 amu. Find the percent abundance of each isotope. (3 points)

Work:

(d) Complete the fusion equation by writing the missing symbol with its mass number and atomic number. (1 point)

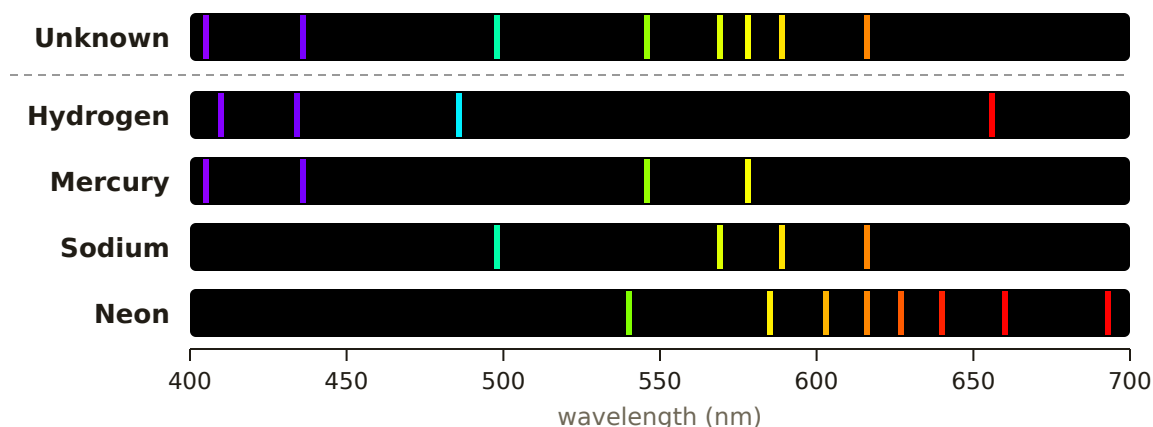


21. Light and spectra: use the correct key terms (8 points) Unit 1

(a) Hydrogen gas in a tube glows when electricity passes through it, and a red line appears in its spectrum. Describe, in order, what happens to an electron in a hydrogen atom. Use all of these terms: *ground state*, *absorbs*, *excited state*, *emits*, *photon*. (3 points)

(b) Explain why hydrogen gives a few separate lines and not a continuous rainbow. (2 points)

(c) Which of the four reference elements are in the unknown? For each element you reject, name the evidence. (3 points)



22. Periodic trends: explain with nuclear charge, shielding and energy levels (8 points) Unit 2

(a) Arrange K, Cl, Na and S in order of increasing first ionization energy. (2 points)

(b) Fluorine has more electrons than lithium, yet a fluorine atom is smaller. Explain. (2 points)

(c) Rank S^{2-} , Cl^{-} , K^{+} and Ca^{2+} from largest to smallest, and explain your order. (2 points)

(d) The first four ionization energies of element X are 738, 1451, 7733 and 10 540 kJ/mol. State the number of valence electrons in X and the charge of its most common ion. Explain how the data shows this. (2 points)

23. Chemical bonding (10 points) Unit 3

(a) For each pair of atoms, calculate ΔEN and state the type of bond. (2 points)

Atoms	ΔEN (show the subtraction)	Type of bond
Mg and O		
C and Cl		

(b) Complete the table. (4 points)

Name	Formula
aluminum oxide	
iron(II) nitrate	
	P_2O_5
	$(NH_4)_2SO_4$

(c) CO₂ and H₂O both contain polar bonds. Draw the Lewis structure of each, showing every lone pair, and explain why only one of the two molecules is polar. (4 points)

CO₂:

H₂O: